Shakil Ahmed Rafi

Ph.D. Candidate in Applied Mathematics
Aspiring Data Scientist & M.L. Engineer
191-4 South Hill Ave., Fayetteville, AR, 72701

mailto:sarafi@uark.edu
+1 770 686 7234
https://shorturl.at/abgmp
https://GitHub.com/2shakilrafi
https://2shakilrafi.github.io

Passionate and hard-working graduate student working at the intersection of machine learning, deep learning, differential equations, and public policy. I aim to combine the rigor of mathematics, the communication skills of a community leader, and the empathy of a researcher in policy to the table.

Education

University of Arkansas, Ph.D.
Applied Mathematics. *Jan, 2020 —May, 2024, expected.*
Recipient, Lawrence Jesser Toll Jr. Endowed Scholarship.
Treasurer, Graduate Student Colloquium.
Fmr. Vice President, Bangladesh Student Organization.

University of Arkansas, M.Sc.
Pure Mathematics. *Aug 2016- Dec 2019.*
Recipient, Bangladesh-Sweden Travel Fund.

Troy University, B.Sc.
Pure Mathematics & Philosophy.
Jan 2011—May 2015.
Graduated Cum Laude
Member, Alabama Eta Chapter of Pi Mu Epsilon, the National Math Honor Society.
Recipient, Millenium Scholar's Award.

Employment History

Teaching Assistant, Department of Data Science Sam M. Walton College of Business, University of Arkansas *Jan 2023 —Present*
Assisting a class on Optimization Methods, and a course on Introduction to Data Science Methods.

Intern, Health Economics, Arkansas Blue Cross & Blue Shield
May 2023 —August 2023
Used Python to model birth and diabetes outcomes as a Markov Model. Trained neural nets in `Keras`/`Tensorflow`to predict birth outcomes from pre-existing conditions.

Graduate Teaching Assistant & Ph.D. Candidate, University of Arkansas
Aug 2016 —Dec 2022
Responsible for teaching, lesson-planning, and interfacing with a cohort of faculty in teaching 500+ students along with grading and office hours.

Asst. Business Analyst, R.S.S. Wears Ltd.
Jul 2015 —Jul 2016
Implemented software for a garment manufacturing start-up, including, payroll, book- keeping with Excel, website setup, paper-trail, and staff directories.

Technical Skills

Author of the `R` package `nnR`.

Reviewer at the Journal of Open Source Software.

Skilled in base-R, R-Studio, tidyverse, Posit Workbench, and Shiny.

Skilled in Python, including: Scikit-Learn, XGBoost, Pandas, Folium, GeoPandas, MatPlotLib, Keras, NumPy, PyTorch, TensorFlow, CVXPY, Pyomo, and other tools.

Proficient in the use LLM models such as Llama and generative AI such as Stable Diffusion, and building LLMs for domain specific tasks.

Proficient in SQL, NoSQL, Spark and Hadoop.

Certified Wolfram Technology Associate.

Skilled in `MATLAB`.

Proficient in `Julia`.

Proficient in PowerBI.

Papers & Preprints

Towards an Algebraic Framework For Approximating Functions Using Neural Network Polynomials—(2024) Rafi, Padgett & Nakarmi

We propose neural network polynomials as a way of approximating certain functions. DOI: `https://doi.org/10.48550/arXiv.2402.01058`
Submitted to ICML 2024.

Gender Disparities in Arkansas, and Income Disparities in the US for the PPP Loan Program —(2023) Rafi

We show that adjusting for business size and median income of zip codes, there seems to be less PPP loans going to black and women owned businesses in Arkansas. DOI: `https://doi.org/10.20944/preprints202309.0654.v1`

Who rides Uber anyway? A census-tract level analysis and clustering of ride-shares for the city of Chicago during the era of the pandemic —(2022) Rafi & Nithila

We show that during the pandemic, the interaction of poorer neighborhoods with ride-sharing infrastructure differs markedly from those of richer neighborhoods. DOI: `https://doi.org/10.36227/techrxiv.21076042.v2`.
Submitted to Springer-Nature's *Transportation*

Employment History — contd.

Concessions Manager, Sodexo Inc.
Aug 2011 —May 2013
Set up concessions for Troy University games, with a team of six other people, including logistics, safety and customer satisfaction.

Certifications

NVIDIA Deep Learning Institute Fundamentals of Deep Learning
NVIDIAInc. Oct 2023
A certification on the use of NVIDIA tools for the most recent deep learning methodologies.

IBM Data Science Professional
IBM with Coursera, Inc. Dec 2021
A certification issued by IBM in collaboration with Coursera, Inc. showing the fundamentals of data science as well as culminating in a real-life project predicting the efficacy of SpaceX launches.

Advanced Google Analytics
Google, Inc. May 2019
Advanced tools and techniques of the Google Analytics platform.

Certified Wolfram Technology Associate
Wolfram Technologies, Inc. Jan 2015
Fundamentals of Wolfram Mathematica and the Wolfram Cloud Platform.

Talks Given

Some parameter estimates for neural network approximants of e^x, the diffusion equation, and modified trapezoidal methods
SIAM Conference on Uncertainty Quantification '24. Trieste, Italy.
We present a novel way of estimating the number of parameters in a feed-forward neural network using purely algebraic operations.

SOM-where in Chicago
Fall 2023 OAK (Oklahoma-Arkansas-Kansas) Supercomputing Conference, Little Rock, AR
We analyzed self-organized maps as way to cluster ride-sharing customers in Chicago during the pandemic.

Machine Learning Through the Centuries
SIAM Applied Mathematics Seminar
Fall 2023
Broad overview of humanity's efforts to understand the world through statistics and machine learning, and the current challenges of ethics and equitability in algorithmic systems.

K-means clustering analysis of ridesharing behaviors in Chicago during the pandemic.
SIAM Mathematics of Data Science Conference, San Diego, CA
Fall 2022
We explore K-means as away to gain better understanding of ridesharing consumer behavior.

Workshops

Mathematical Sciences Research Institute Summer Graduate School in Machine Learning, 2023, San Diego, CA
Jun 2023 —Jul 2023
Nominated by my department chair, and went on to attend the prestigious MSRI Summer Graduate School on Machine Learning. We attended and implemented a two week workshop on overparametrization and adversial patterns in image recognition.
Funded by
Arkansas Summer Research Institute, 2022, Online
May 2022 —June 2022
We looked at loans disbursed during the pandemic (PPP) and saw that poorer neighborhoods tended to get smaller and less frequent loans.
Funded by NSF and the University of Arkansas.

Service

Lead Judge, Computer Science Category, Northwest Arkansas Regional Science and Engineering Fair
2023, 2024
Served as the Lead Judge in judging over two dozen, very competitive projects involving image recognition, neural networks, and basic NLP.

Vice President of the Bangladesh Student Organization
Aug 2019 – May 2020
Served in an organization dedicated to disseminating Bangladeshi culture and helping over a hundred students of Bangladeshi origin on campus to thrive and feel at home.

Departmental Representative, Graduate Student Council
May 2021 – May 2022
Representing the interests of the fifty graduate students of the Department of Mathematical Sciences at the Fulbright College of Arts and Sciences.

Treasurer, Graduate Student Colloquium
Aug 2022 – May 2023
Responsible for securing funding for speakers when they may appear at our weekly and bi-weekly talks at the Department of Mathematical Sciences.